Bash Scripting Suite for System Maintenance

Project Structure :

```
maintenance-suite/
|
├── backups/                 → Backup storage
├── logs/                    → All script logs
├── scripts/
|   ├── backup.sh            → Automates incremental backups
|   ├── update_cleanup.sh    → Updates packages & cleans cache
|   ├── log_monitor.sh       → Monitors logs for warnings/errors
|   ├── maintenance_menu.sh → Interactive menu to run all tasks
|   └── lib_logger.sh        → Shared logging functions
```

Day-wise Breakdown

Day	Script	Purpose
Day 1	backup.sh	Uses rsync to perform incremental backups. Configurable sources and automatic pruning of old backups.
Day 2	update_cleanup.sh	Updates the system via apt/dnf, removes unused packages, and cleans cache.
Day 3	log_monitor.sh	Scans /var/log/auth.log (or other logs) for “Failed password”, “error”, etc., and records alerts.
Day 4	maintenance_menu.sh	Provides a CLI menu to run backup, update, log monitor, or all tasks together.
Day 5	lib_logger.sh + Error Traps	Adds centralized logging and error handling for reliability.

Project Structure :

The project is divided into **five main scripts**, stored in the folder `maintenance-suite/scripts`.

1. **`backup.sh`** —
   This script creates **incremental backups** of important directories using `rsync`.
   It automatically adds timestamps, logs every step, and deletes backups older than 14 days to save space.

2. **`update_cleanup.sh`** —
   This script automates **system updates and cleanups** using `apt`.
   It runs `apt update`, `apt upgrade`, `autoremove`, and `autoclean` — making system maintenance one click.

3. **`log_monitor.sh`** —
   This script checks `/var/log/auth.log` for **failed logins or system errors**.
   It highlights suspicious entries like "Failed password" or "error" and saves the results to a log file.

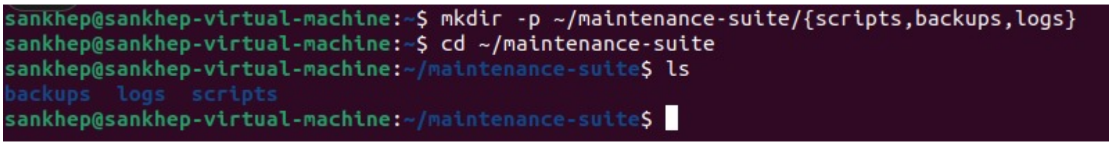
4. **`lib_logger.sh`** —
   A **shared library** that all scripts use for logging.
   It adds timestamps, error traps, and maintains a **central master log** for all activities.

5. **`maintenance_menu.sh`** —
   This is the **main controller script**.
   It shows a user-friendly menu with options to run backup, update, log monitor, or all tasks together.
   Each selection runs the corresponding script with root permissions if required.

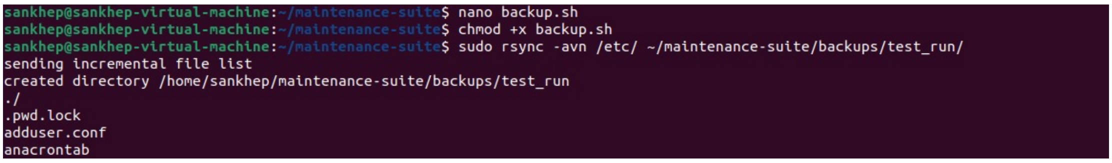
DAY 1

Automated System Backup Script (backup.sh)

To build a Bash script that automatically performs **incremental backups** of important system directories using rsync, stores them safely, and logs all activities.

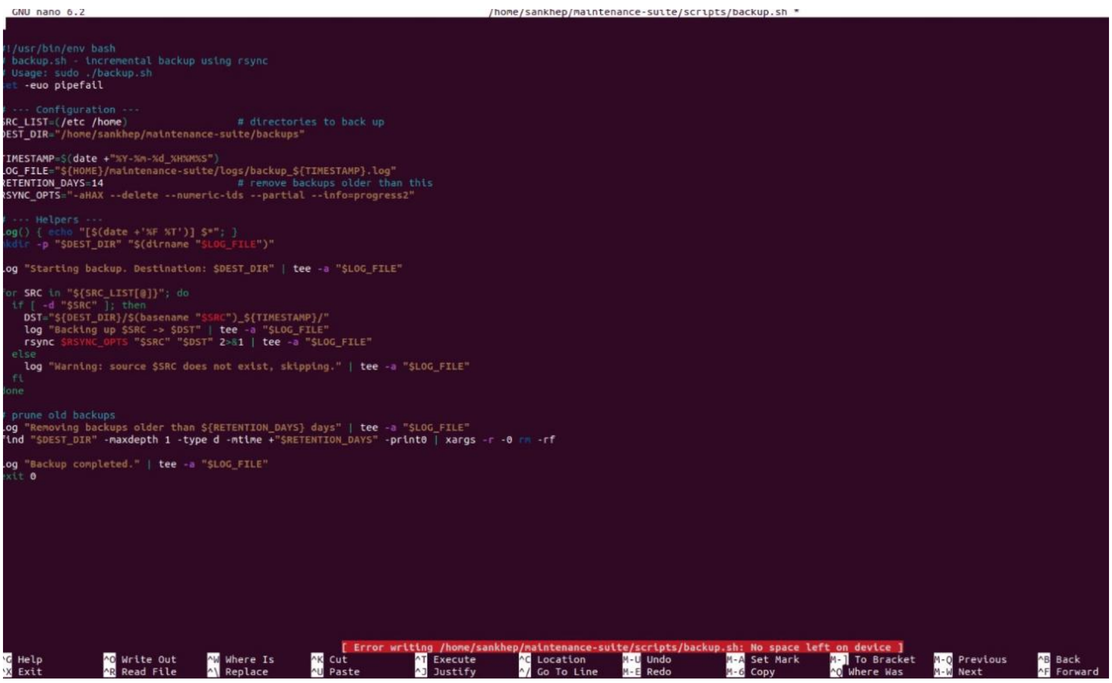


```
sankhep@sankhep-virtual-machine:~$ mkdir -p ~/maintenance-suite/{scripts,backups,logs}
sankhep@sankhep-virtual-machine:~$ cd ~/maintenance-suite
sankhep@sankhep-virtual-machine:~/maintenance-suite$ ls
backups  logs  scripts
sankhep@sankhep-virtual-machine:~/maintenance-suite$
```



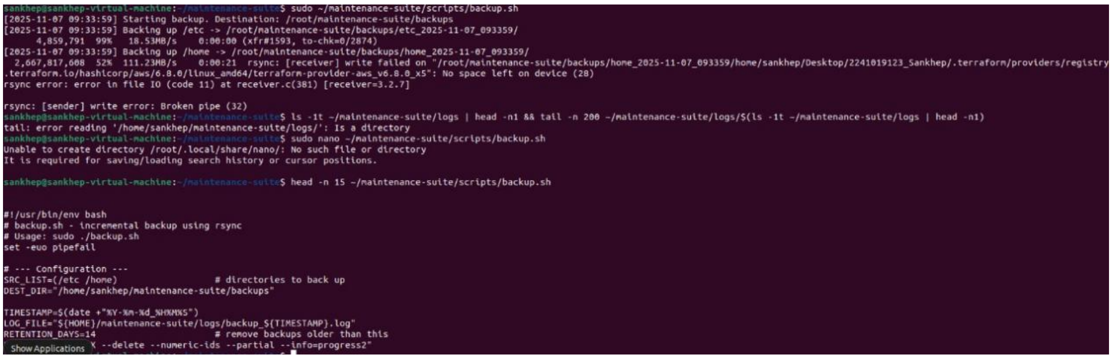
```
sankhep@sankhep-virtual-machine:~/maintenance-suite$ nano backup.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ chmod +x backup.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ sudo rsync -avn /etc/ ~/maintenance-suite/backups/test_run/
sending incremental file list
created directory /home/sankhep/maintenance-suite/backups/test_run
./
.pwd.lock
adduser.conf
anacrontab
```

CODE :



```
GNU nano 6.2                                   /home/sankhep/maintenance-suite/scripts/backup.sh *

!/usr/bin/env bash
 backup.sh - incremental backup using rsync
 Usage: sudo ./backup.sh
et -euo pipefail

 --- Configuration ---
RC_LIST=(/etc /home)                          # directories to back up
EST_DIR="/home/sankhep/maintenance-suite/backups"

IMESTAMP=$(date +"%Y-%m-%d_%H%M%S")
OG_FILE="${HOME}/maintenance-suite/logs/backup_${TIMESTAMP}.log"
ETENTION_DAYS=14                              # remove backups older than this
SYNC_OPTS="-aHAX --delete --numeric-ids --partial --info=progress2"

 --- Helpers ---
og() { echo "[$(date +'%F %T')] $*"; }
kdir -p "$DEST_DIR" "$(dirname "$LOG_FILE")"

og "Starting backup. Destination: $DEST_DIR" | tee -a "$LOG_FILE"

or SRC in "${SRC_LIST[@]}"; do
 if [ -d "$SRC" ]; then
   DST="${DEST_DIR}/$(basename "$SRC")_${TIMESTAMP}/"
   log "Backing up $SRC -> $DST" | tee -a "$LOG_FILE"
   rsync $RSYNC_OPTS "$SRC" "$DST" 2>&1 | tee -a "$LOG_FILE"
 else
   log "Warning: source $SRC does not exist, skipping." | tee -a "$LOG_FILE"
 fi
lone

 prune old backups
og "Removing backups older than ${RETENTION_DAYS} days" | tee -a "$LOG_FILE"
ind "$DEST_DIR" -maxdepth 1 -type d -mtime +"$RETENTION_DAYS" -print0 | xargs -r -0 rm -rf

og "Backup completed." | tee -a "$LOG_FILE"
xit 0

                    [ Error writing /home/sankhep/maintenance-suite/scripts/backup.sh: No space left on device ]
^G Help      ^O Write Out  ^W Where Is  ^K Cut    ^T Execute  ^C Location    M-U Undo  M-A Set Mark  M-] To Bracket  M-Q Previous  ^B Back
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line  M-E Redo  M-6 Copy      ^Q Where Was    M-W Next      ^F Forward
```

OUTPUT :



You created a script named:

~/maintenance-suite/scripts/backup.sh

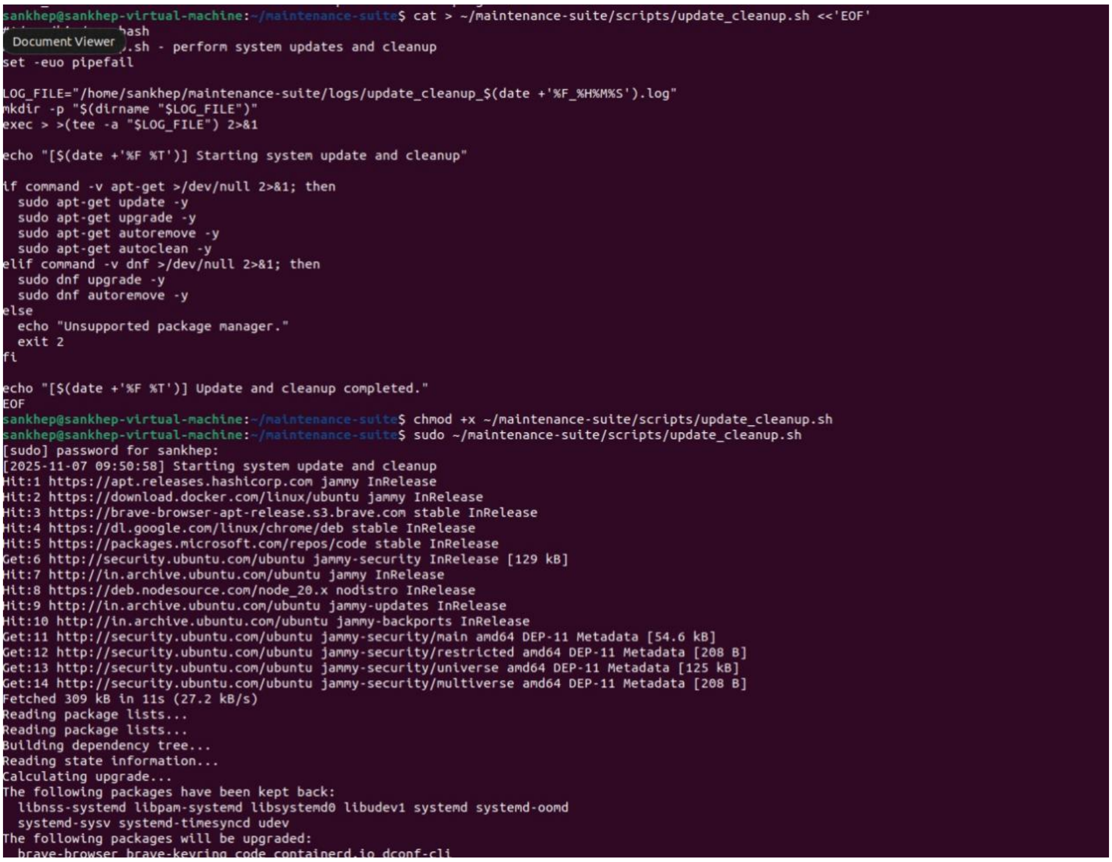
This script:

- **Backs up important directories** — like `/etc` and `/home`.
- **Uses** `rsync` to copy only changed files (efficient incremental backup).
- **Adds timestamps** to every backup folder.
- **Stores logs** in `~/maintenance-suite/logs/`.
- **Automatically deletes old backups** (older than 14 days).

DAY 2

System Update & Cleanup Script (update_cleanup.sh) :

To automate system updates, upgrades, and cleanup tasks using a Bash script — ensuring your Linux system stays secure and optimized without manual commands.



```
sankhep@sankhep-virtual-machine:~/maintenance-suite$ cat > ~/maintenance-suite/scripts/update_cleanup.sh <<'EOF'
bash
Document Viewer .sh - perform system updates and cleanup
set -euo pipefail

LOG_FILE="/home/sankhep/maintenance-suite/logs/update_cleanup_$(date +'%F_%H%M%S').log"
mkdir -p "$(dirname "$LOG_FILE")"
exec > >(tee -a "$LOG_FILE") 2>&1

echo "[$(date +'%F %T')] Starting system update and cleanup"

if command -v apt-get >/dev/null 2>&1; then
  sudo apt-get update -y
  sudo apt-get upgrade -y
  sudo apt-get autoremove -y
  sudo apt-get autoclean -y
elif command -v dnf >/dev/null 2>&1; then
  sudo dnf upgrade -y
  sudo dnf autoremove -y
else
  echo "Unsupported package manager."
  exit 2
fi

echo "[$(date +'%F %T')] Update and cleanup completed."
EOF
sankhep@sankhep-virtual-machine:~/maintenance-suite$ chmod +x ~/maintenance-suite/scripts/update_cleanup.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ sudo ~/maintenance-suite/scripts/update_cleanup.sh
[sudo] password for sankhep:
[2025-11-07 09:50:58] Starting system update and cleanup
Hit:1 https://apt.releases.hashicorp.com jammy InRelease
Hit:2 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:3 https://brave-browser-apt-release.s3.brave.com stable InRelease
Hit:4 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:5 https://packages.microsoft.com/repos/code stable InRelease
Get:6 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:7 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:8 https://deb.nodesource.com/node_20.x nodistro InRelease
Hit:9 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:10 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main amd64 DEP-11 Metadata [54.6 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 DEP-11 Metadata [208 B]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 DEP-11 Metadata [125 kB]
Get:14 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 DEP-11 Metadata [208 B]
Fetched 309 kB in 11s (27.2 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
The following packages have been kept back:
  libnss-systemd libpam-systemd libsystemd0 libudev1 systemd systemd-oomd
  systemd-sysv systemd-timesyncd udev
The following packages will be upgraded:
  brave-browser brave-keyring code containerd.io dconf-cli
```

You created the script:
~/maintenance-suite/scripts/update_cleanup.sh
This script:

- **Automatically updates** the system's package lists.
- **Upgrades** all outdated packages.
- **Removes unnecessary packages** with `autoremove`.
- **Cleans up package caches** with `autoclean`.
- **Logs all activities** to a timestamped log file.

DAY 3

Log Monitoring Script (log_monitor.sh):

To build a Bash script that automatically **scans system logs** for common errors, failed login attempts, or warnings — helping identify potential issues or security breaches.

Code :

```
  GNU nano 6.2                                   /home/sankhep/maintenance-suite/scripts/log_monitor.sh
#!/usr/bin/env bash
# log_monitor.sh - checks system logs for warnings or failures
set -euo pipefail

LOG_FILE="/home/sankhep/maintenance-suite/logs/log_monitor_$(date +'%F_%H%M%S').log"
mkdir -p "$(dirname "$LOG_FILE")"

TARGET_LOG=${1:-/var/log/auth.log}
PATTERN=${2:-"Failed password|error|CRON"}

echo "[$(date +'%F %T')] Scanning $TARGET_LOG for pattern: $PATTERN" | tee -a "$LOG_FILE"

if [ ! -r "$TARGET_LOG" ]; then
  echo "Cannot read $TARGET_LOG (try with sudo)" | tee -a "$LOG_FILE"
  exit 1
fi

tail -n 500 "$TARGET_LOG" | grep -E -n "$PATTERN" | tee -a "$LOG_FILE" || \
  echo "No matches found in last 500 lines." | tee -a "$LOG_FILE"

echo "[$(date +'%F %T')] Log monitor finished." | tee -a "$LOG_FILE"
```

Output :

```
sankhep@sankhep-virtual-machine:~/maintenance-suite$ nano ~/maintenance-suite/scripts/log_monitor.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ chmod +x ~/maintenance-suite/scripts/log_monitor.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ sudo ~/maintenance-suite/scripts/log_monitor.sh
[2025-11-07 09:58:26] Scanning /var/log/auth.log for pattern: Failed password|error|CRON
44:Nov  7 09:17:01 sankhep-virtual-machine CRON[6092]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
45:Nov  7 09:17:01 sankhep-virtual-machine CRON[6092]: pam_unix(cron:session): session closed for user root
46:Nov  7 09:30:01 sankhep-virtual-machine CRON[6271]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
47:Nov  7 09:30:01 sankhep-virtual-machine CRON[6271]: pam_unix(cron:session): session closed for user root
[2025-11-07 09:58:26] Log monitor finished.
sankhep@sankhep-virtual-machine:~/maintenance-suite$ ls -lt ~/maintenance-suite/logs | head -n3
total 8
-rw-r--r-- 1 root root  601 Nov  7 09:58 log_monitor_2025-11-07_095826.log
-rw-r--r-- 1 root root 4020 Nov  7 09:51 update_cleanup_2025-11-07_095058.log
sankhep@sankhep-virtual-machine:~/maintenance-suite$
```

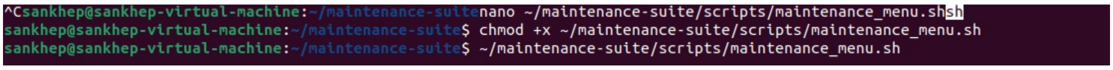
You created the script:
~/maintenance-suite/scripts/log_monitor.sh
This script:

Reads the system log file (default: `/var/log/auth.log`).
Searches for **specific patterns** like “Failed password”, “error”, or “CRON”
Prints matching log lines with line numbers.
Logs the results into a separate timestamped file.
Displays “No matches found” if there are no warnings.

DAY 4

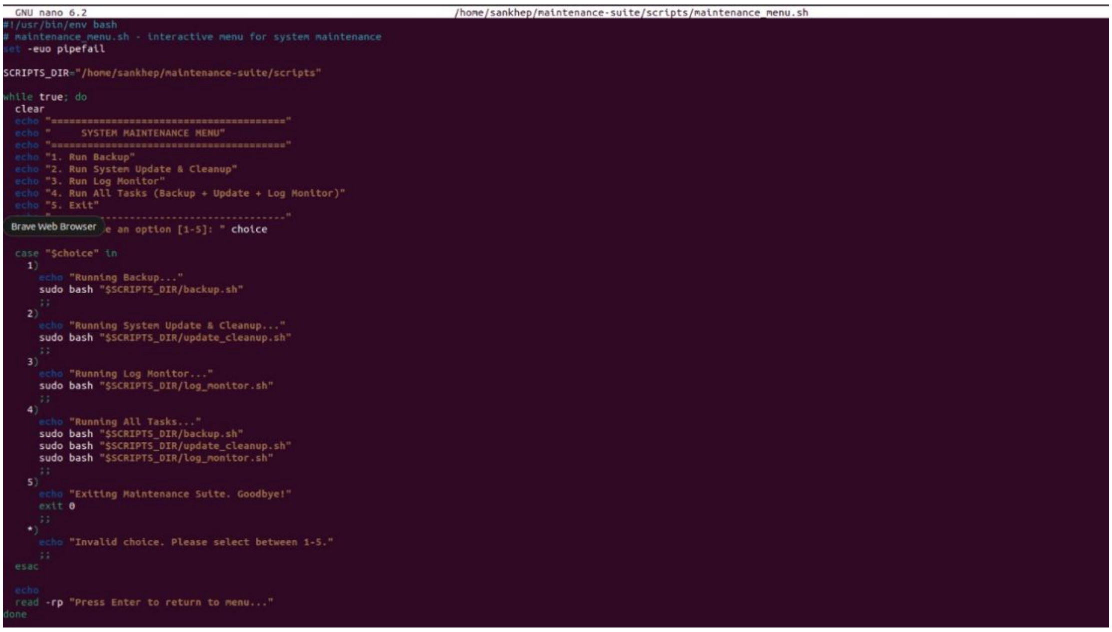
Maintenance Menu Script (maintenance_menu.sh):

To design a single **interactive Bash menu** that integrates all your scripts —
backup.sh, update_cleanup.sh, and log_monitor.sh — into one unified system maintenance tool.



```
^Csankhep@sankhep-virtual-machine:~/maintenance-suitenano ~/maintenance-suite/scripts/maintenance_menu.shsh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ chmod +x ~/maintenance-suite/scripts/maintenance_menu.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ ~/maintenance-suite/scripts/maintenance_menu.sh
```

CODE :



```
GNU nano 6.2                    /home/sankhep/maintenance-suite/scripts/maintenance_menu.sh
#!/usr/bin/env bash
# maintenance_menu.sh - interactive menu for system maintenance
set -euo pipefail

SCRIPTS_DIR="/home/sankhep/maintenance-suite/scripts"

while true; do
  clear
  echo "========================================="
  echo "      SYSTEM MAINTENANCE MENU"
  echo "========================================="
  echo "1. Run Backup"
  echo "2. Run System Update & Cleanup"
  echo "3. Run Log Monitor"
  echo "4. Run All Tasks (Backup + Update + Log Monitor)"
  echo "5. Exit"
  echo "-----------------------------------------"
Brave Web Browser e an option [1-5]: " choice

  case "$choice" in
    1)
      echo "Running Backup..."
      sudo bash "$SCRIPTS_DIR/backup.sh"
      ;;
    2)
      echo "Running System Update & Cleanup..."
      sudo bash "$SCRIPTS_DIR/update_cleanup.sh"
      ;;
    3)
      echo "Running Log Monitor..."
      sudo bash "$SCRIPTS_DIR/log_monitor.sh"
      ;;
    4)
      echo "Running All Tasks..."
      sudo bash "$SCRIPTS_DIR/backup.sh"
      sudo bash "$SCRIPTS_DIR/update_cleanup.sh"
      sudo bash "$SCRIPTS_DIR/log_monitor.sh"
      ;;
    5)
      echo "Exiting Maintenance Suite. Goodbye!"
      exit 0
      ;;
    *)
      echo "Invalid choice. Please select between 1-5."
      ;;
  esac

  echo
  read -rp "Press Enter to return to menu..."
done
```

You created the main controller script:

~/maintenance-suite/scripts/maintenance_menu.sh

This script:

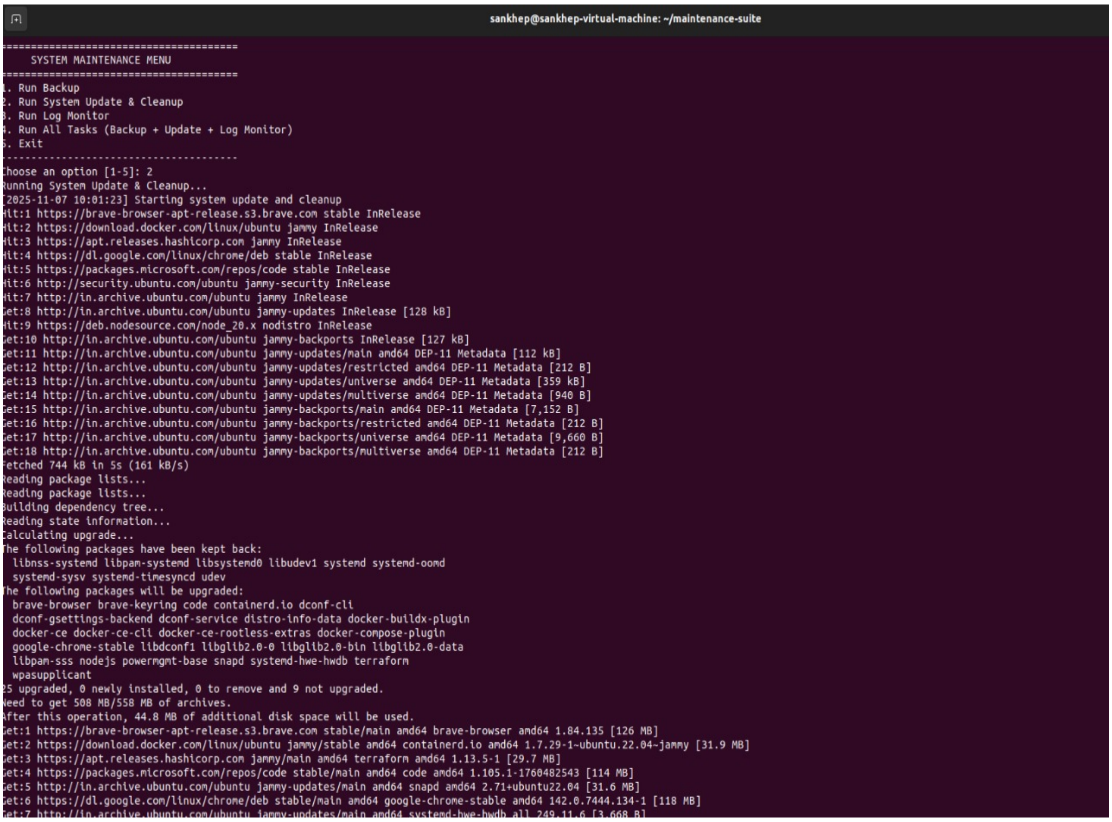
Displays an **interactive text menu** in the terminal.

Lets users choose to run **any one task** or **all tasks together**.

Automatically executes the corresponding scripts with `sudo` when required.

Acts as the **central hub** of your 5-day maintenance suite.

OUTPUT :



```
sankhep@sankhep-virtual-machine: ~/maintenance-suite
======================================
    SYSTEM MAINTENANCE MENU
======================================
1. Run Backup
2. Run System Update & Cleanup
3. Run Log Monitor
4. Run All Tasks (Backup + Update + Log Monitor)
5. Exit
--------------------------------------
Choose an option [1-5]: 2
Running System Update & Cleanup...
[2025-11-07 10:01:23] Starting system update and cleanup
Hit:1 https://brave-browser-apt-release.s3.brave.com stable InRelease
Hit:2 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:3 https://apt.releases.hashicorp.com jammy InRelease
Hit:4 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:5 https://packages.microsoft.com/repos/code stable InRelease
Hit:6 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:7 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Hit:9 https://deb.nodesource.com/node_20.x nodistro InRelease
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 DEP-11 Metadata [112 kB]
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 DEP-11 Metadata [212 B]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 DEP-11 Metadata [359 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 DEP-11 Metadata [940 B]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-backports/main amd64 DEP-11 Metadata [7,152 B]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 DEP-11 Metadata [212 B]
Get:17 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 DEP-11 Metadata [9,660 B]
Get:18 http://in.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 DEP-11 Metadata [212 B]
Fetched 744 kB in 5s (161 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
The following packages have been kept back:
  libnss-systemd libpam-systemd libsystemd0 libudev1 systemd systemd-oomd
  systemd-sysv systemd-timesyncd udev
The following packages will be upgraded:
  brave-browser brave-keyring code containerd.io dconf-cli
  dconf-gsettings-backend dconf-service distro-info-data docker-buildx-plugin
  docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin
  google-chrome-stable libdconf1 libglib2.0-0 libglib2.0-bin libglib2.0-data
  libpam-sss nodejs powermgmt-base snapd systemd-hwe-hwdb terraform
  wpasupplicant
25 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
Need to get 508 MB/558 MB of archives.
After this operation, 44.8 MB of additional disk space will be used.
Get:1 https://brave-browser-apt-release.s3.brave.com stable/main amd64 brave-browser amd64 1.84.135 [126 MB]
Get:2 https://download.docker.com/linux/ubuntu jammy/stable amd64 containerd.io amd64 1.7.29-1~ubuntu.22.04~jammy [31.9 MB]
Get:3 https://apt.releases.hashicorp.com jammy/main amd64 terraform amd64 1.13.5-1 [29.7 MB]
Get:4 https://packages.microsoft.com/repos/code stable/main amd64 code amd64 1.105.1-1760482543 [114 MB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 snapd amd64 2.71+ubuntu22.04 [31.6 MB]
Get:6 https://dl.google.com/linux/chrome/deb stable/main amd64 google-chrome-stable amd64 142.0.7444.134-1 [118 MB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 systemd-hwe-hwdb all 249.11.6 [3,668 B]
```

MENU BAR :

```
==========================================
     SYSTEM MAINTENANCE MENU
==========================================
1. Run Backup
2. Run System Update & Cleanup
3. Run Log Monitor
4. Run All Tasks (Backup + Update + Log Monitor)
5. Exit
------------------------------------------
Choose an option [1-5]:
```

DAY 5

Error Handling & Centralized Logging (lib_logger.sh):

To add **robust error handling**, **centralized logging**, and a **shared logger library** (lib_logger.sh) so all scripts in your maintenance suite are reliable, consistent, and easy to debug.

CODE :

```
  GNU nano 6.2                                            /home/sankhep/maintenance-suite/scripts/lib_logger.sh *
#!/usr/bin/env bash
# lib_logger.sh - shared logger library

LOG_DIR="/home/sankhep/maintenance-suite/logs"
MASTER_LOG="$LOG_DIR/maintenance_master.log"
mkdir -p "$LOG_DIR"

_log() {
  local level="$1"; shift
  echo "[$(date +'%F %T')] [$level] $*" | tee -a "$MASTER_LOG"
}

info()  { _log INFO  "$@"; }
warn()  { _log WARN  "$@"; }
error() { _log ERROR "$@"; }
```

```
sankhep@sankhep-virtual-machine:~/maintenance-suite$ sudo nano ~/maintenance-suite/scripts/backup.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ sudo nano ~/maintenance-suite/scripts/update_cleanup.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ sudo nano ~/maintenance-suite/scripts/log_monitor.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$ ~/maintenance-suite/scripts/maintenance_menu.sh
```

Additionals :

You added these 3 lines right below set -euo pipefail in each of your main scripts:

```
trap 'error "Script aborted or failed at line $LINENO"; exit 1' ERR INT TERM
source /home/sankhep/maintenance-suite/scripts/lib_logger.sh
info "Starting $(basename "$0")"
```

OUTPUT :

```
========================================
     SYSTEM MAINTENANCE MENU
========================================
1. Run Backup
2. Run System Update & Cleanup
3. Run Log Monitor
4. Run All Tasks (Backup + Update + Log Monitor)
5. Exit
----------------------------------------
Choose an option [1-5]: 4
Running All Tasks...
[2025-11-07 10:19:00] [INFO] Starting backup.sh
[2025-11-07 10:19:00] Starting backup. Destination: /home/sankhep/maintenance-suite/backups
[2025-11-07 10:19:00] Backing up /etc -> /home/sankhep/maintenance-suite/backups/etc_2025-11-07_101900/
      4,860,083  99%   22.82MB/s    0:00:00 (xfr#1593, to-chk=0/2874)
[2025-11-07 10:19:01] Backing up /home -> /home/sankhep/maintenance-suite/backups/home_2025-11-07_101900/
  8,218,300,584  99%  316.38MB/s    0:00:24 (xfr#59200, to-chk=0/65531)
[2025-11-07 10:19:25] Removing backups older than 14 days
[2025-11-07 10:19:25] Backup completed.
[2025-11-07 10:19:25] [INFO] Starting update_cleanup.sh
[2025-11-07 10:19:25] Starting system update and cleanup
Hit:1 https://brave-browser-apt-release.s3.brave.com stable InRelease
Hit:2 https://download.docker.com/linux/ubuntu jammy InRelease
Get:3 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Hit:4 https://dl.google.com/linux/chrome/deb stable InRelease
Get:5 https://apt.releases.hashicorp.com jammy InRelease [12.9 kB]
Get:6 https://packages.microsoft.com/repos/code stable/main armhf Packages [20.5 kB]
Get:7 https://packages.microsoft.com/repos/code stable/main arm64 Packages [20.5 kB]
Hit:8 http://security.ubuntu.com/ubuntu jammy-security InRelease
Get:9 https://packages.microsoft.com/repos/code stable/main amd64 Packages [20.4 kB]
Hit:10 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:11 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:12 https://deb.nodesource.com/node_20.x nodistro InRelease
Hit:13 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Fetched 77.9 kB in 2s (43.5 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
The following packages have been kept back:
  libnss-systemd libpam-systemd libsystemd0 libudev1 systemd systemd-oomd
  systemd-sysv systemd-timesyncd udev
The following packages will be upgraded:
  brave-browser brave-keyring code containerd.io dconf-cli
  dconf-gsettings-backend dconf-service distro-info-data docker-buildx-plugin
  docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin
  google-chrome-stable libdconf1 libglib2.0-0 libglib2.0-bin libglib2.0-data
  libpam-sss nodejs powermgmt-base snapd systemd-hwe-hwdb terraform
  wpasupplicant
25 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
Need to get 558 MB of archives.
After this operation, 44.8 MB of additional disk space will be used.
Get:1 https://brave-browser-apt-release.s3.brave.com stable/main amd64 brave-keyring all 1.19 [8,116 B]
Get:2 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-cli amd64 5:28.5.2-1~ubuntu.22.04~jammy [16.0 MB]
Get:3 https://apt.releases.hashicorp.com jammy/main amd64 terraform amd64 1.13.5-1 [29.7 MB]
```

```
The following packages have been kept back:
  libnss-systemd libpam-systemd libsystemd0 libudev1 systemd systemd-oomd
  systemd-sysv systemd-timesyncd udev
The following packages will be upgraded:
  brave-browser brave-keyring code containerd.io dconf-cli
Ubuntu 22.04.5 LTS amd64 end dconf-service distro-info-data docker-buildx-plugin
                         cli docker-ce-rootless-extras docker-compose-plugin
  google-chrome-stable libdconf1 libglib2.0-0 libglib2.0-bin libglib2.0-data
  libpam-sss nodejs powermgmt-base snapd systemd-hwe-hwdb terraform
  wpasupplicant
25 upgraded, 0 newly installed, 0 to remove and 9 not upgraded.
Need to get 558 MB of archives.
After this operation, 44.8 MB of additional disk space will be used.
Get:1 https://brave-browser-apt-release.s3.brave.com stable/main amd64 brave-keyring all 1.19 [8,116 B]
Get:2 https://download.docker.com/linux/ubuntu jammy/stable amd64 docker-ce-cli amd64 5:28.5.2-1~ubuntu.22.04~jammy [16.0 MB]
Get:3 https://apt.releases.hashicorp.com jammy/main amd64 terraform amd64 1.13.5-1 [29.7 MB]
Get:4 https://brave-browser-apt-release.s3.brave.com stable/main amd64 brave-browser amd64 1.84.135 [126 MB]
Get:5 https://dl.google.com/linux/chrome/deb stable/main amd64 google-chrome-stable amd64 142.0.7444.134-1 [118 MB]
Get:6 https://packages.microsoft.com/repos/code stable/main amd64 code amd64 1.105.1-1760482543 [114 MB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libglib2.0-data all 2.72.4-0ubuntu2.6 [4,698 B]
Get:8 https://deb.nodesource.com/node_20.x nodistro/main amd64 nodejs amd64 20.19.5-1nodesource1 [32.0 MB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libglib2.0-bin amd64 2.72.4-0ubuntu2.6 [80.9 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libglib2.0-0 amd64 2.72.4-0ubuntu2.6 [1,467 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 distro-info-data all 0.52ubuntu0.11 [5,444 B]
Get:12 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 powermgmt-base all 1.36ubuntu0.22.04.1 [7,736 B]
Get:13 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 dconf-service amd64 0.40.0-3ubuntu0.1 [28.1 kB]
Get:14 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 dconf-gsettings-backend amd64 0.40.0-3ubuntu0.1 [22.7 kB]
Get:15 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 dconf-cli amd64 0.40.0-3ubuntu0.1 [26.5 kB]
Get:16 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libdconf1 amd64 0.40.0-3ubuntu0.1 [40.5 kB]
Get:17 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 snapd amd64 2.71+ubuntu22.04 [31.6 MB]
Get:18 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 systemd-hwe-hwdb all 249.11.6 [3,668 B]
Get:19 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 wpasupplicant amd64 2:2.10-6ubuntu2.3 [1,482 kB]
Get:20 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpam-sss amd64 2.6.3-1ubuntu3.6 [41.0 kB]
Get:21 https://download.docker.com/linux/ubuntu jammy/stable amd64 containerd.io amd64 1.7.29-1~ubuntu.22.04~jammy [31.9 MB]
```

```
Get:20 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpam-sss amd64 2.6.3-1ubuntu3.6 [41.0 kB]
Get:21 https://download.docker.com/linux/ubuntu jammy/stable amd64 containerd.io amd64 1.7.29-1~ubuntu.22.04~jammy [31.9 MB]
^Csankhep@sankhep-virtual-machine:~/maintenance-suitetail -n 30 /home/sankhep/maintenance-suite/logs/maintenance_master.log
[2025-11-07 10:12:32] [INFO] Starting backup.sh
[2025-11-07 10:13:05] [ERROR] Script aborted or failed at line 32
[2025-11-07 10:16:02] [INFO] Starting backup.sh
[2025-11-07 10:16:03] [ERROR] Script aborted or failed at line 33
[2025-11-07 10:19:00] [INFO] Starting backup.sh
[2025-11-07 10:19:25] [INFO] Starting update_cleanup.sh
sankhep@sankhep-virtual-machine:~/maintenance-suite$
```

End of Project Report

Automated System Maintenance Suite (5-Day Bash Project) :

Developer:

Name: Sankhep Sahoo
Course: Computer Science (core)

Project Duration: 5 Days

Project Summary :

The **System Maintenance Suite** automates critical system tasks — backup, update, cleanup, and log monitoring — using Bash scripting.
Each component is modular, tested, and integrated under a unified menu-driven interface.

It ensures:

1. Safe and reliable **system backups**
2. Automated **software updates and cleanup**
3. Real-time **log error detection**
4. Centralized **logging and error reporting**
5. Simple **interactive interface** for all operations

Tools & Technologies :

Language: Bash Shell Scripting

Utilities: `rsync`, `apt`, `grep`, `tee`, `find`, `cron`

Platform: Ubuntu 22.04 LTS

Concepts Used:

1. Automation with Bash
2. File handling & logging
3. Regular expressions
4. Process control (`trap`, `set -euo pipefail`)
5. Menu-based CLI scripting